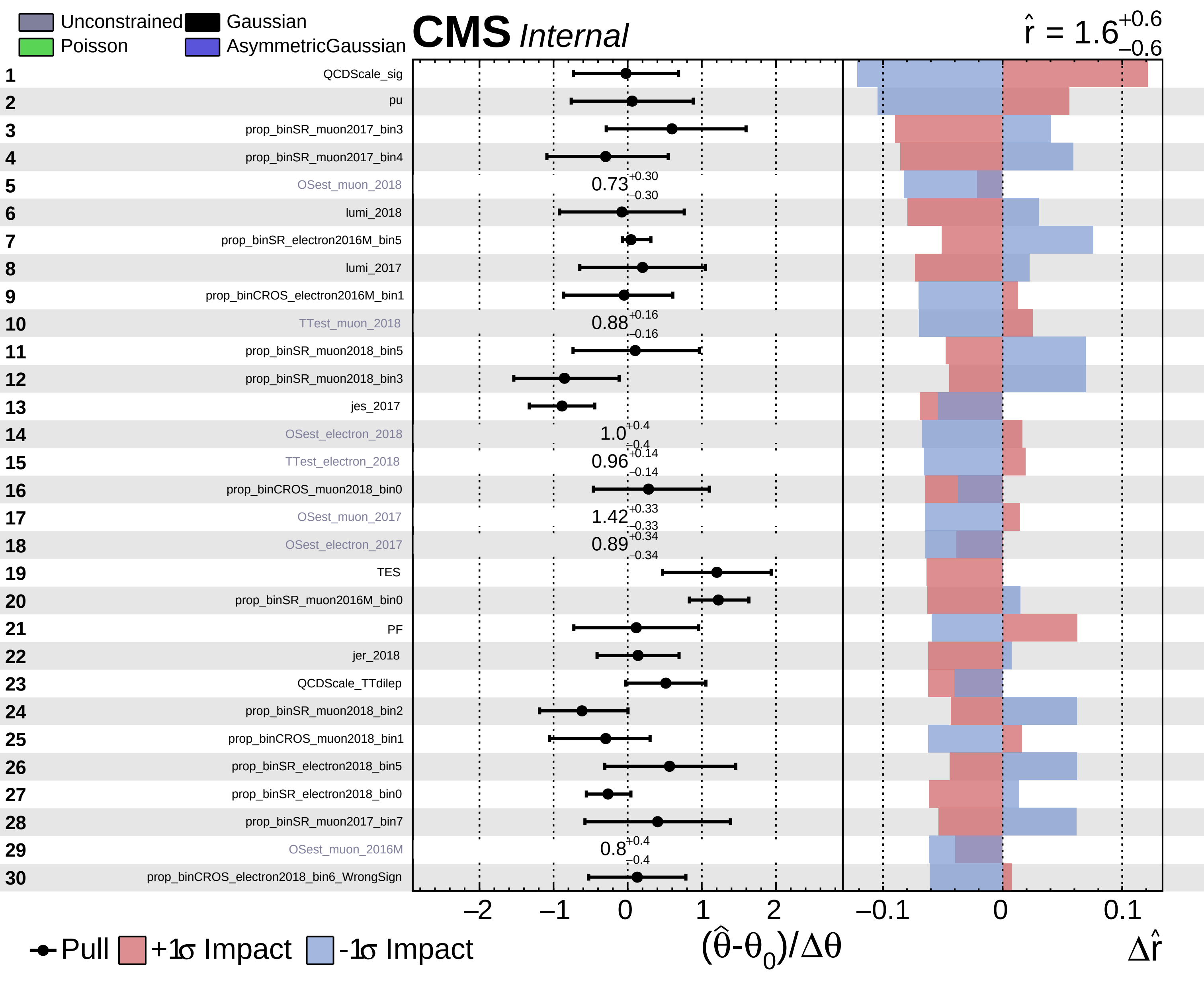
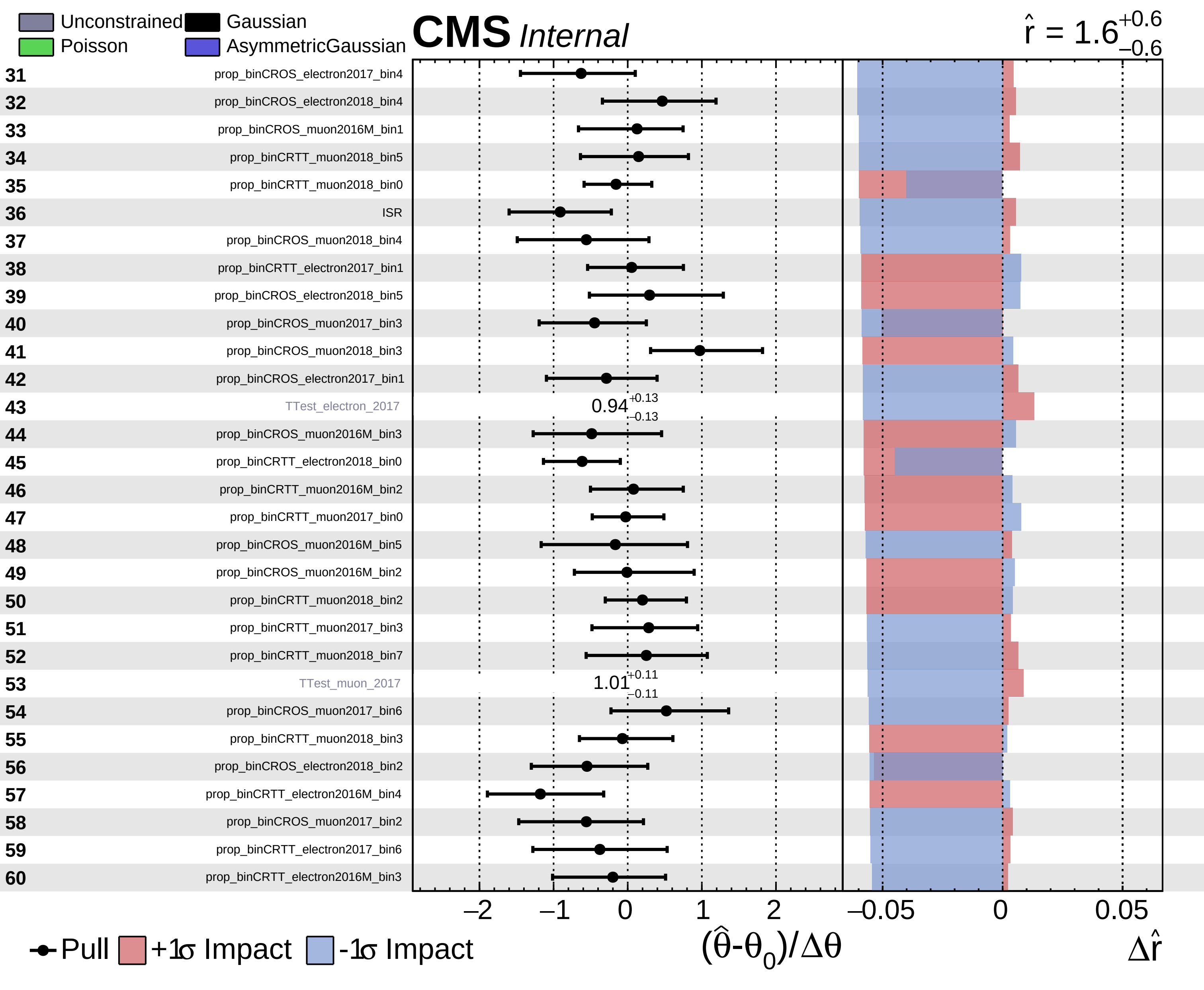
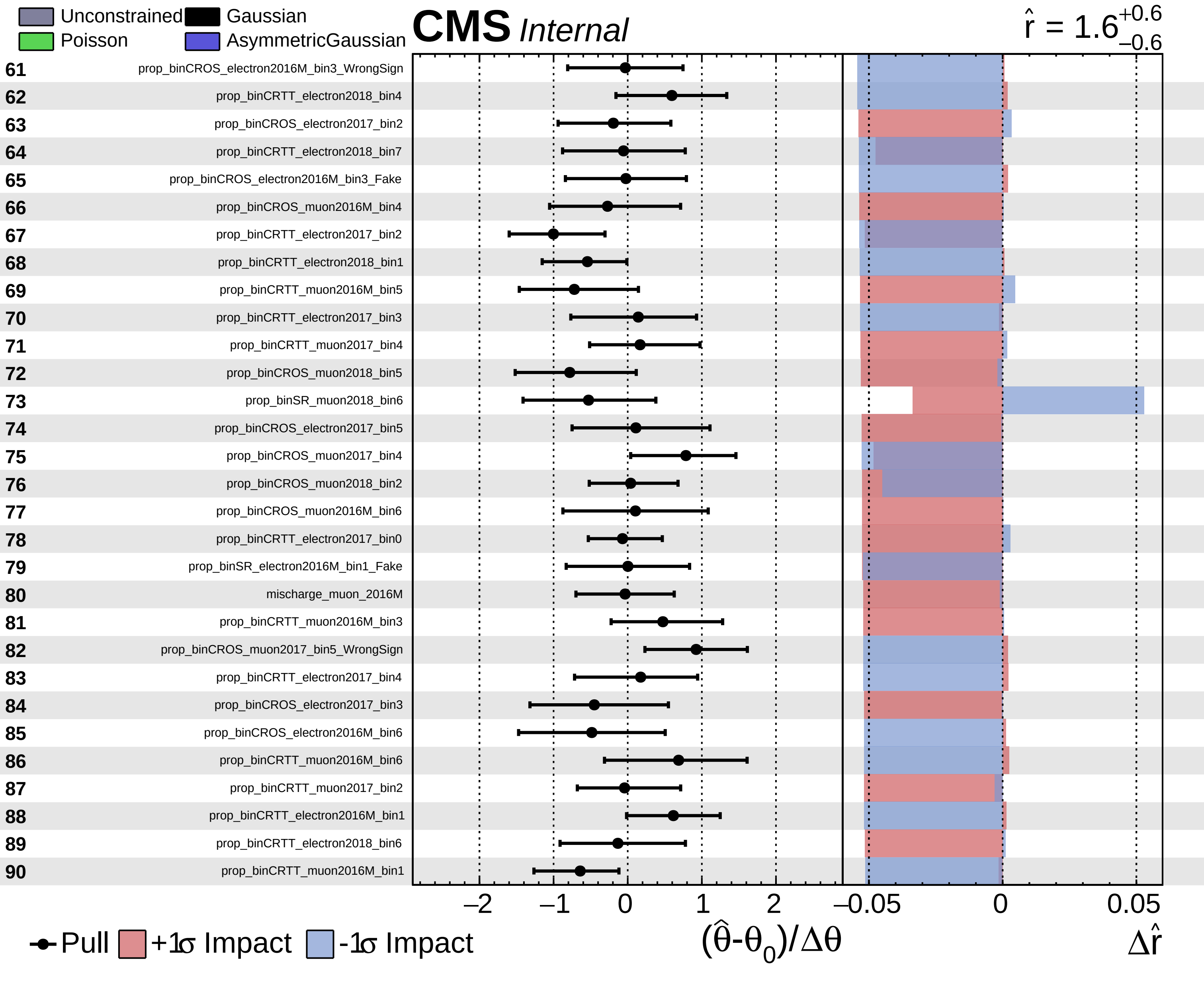
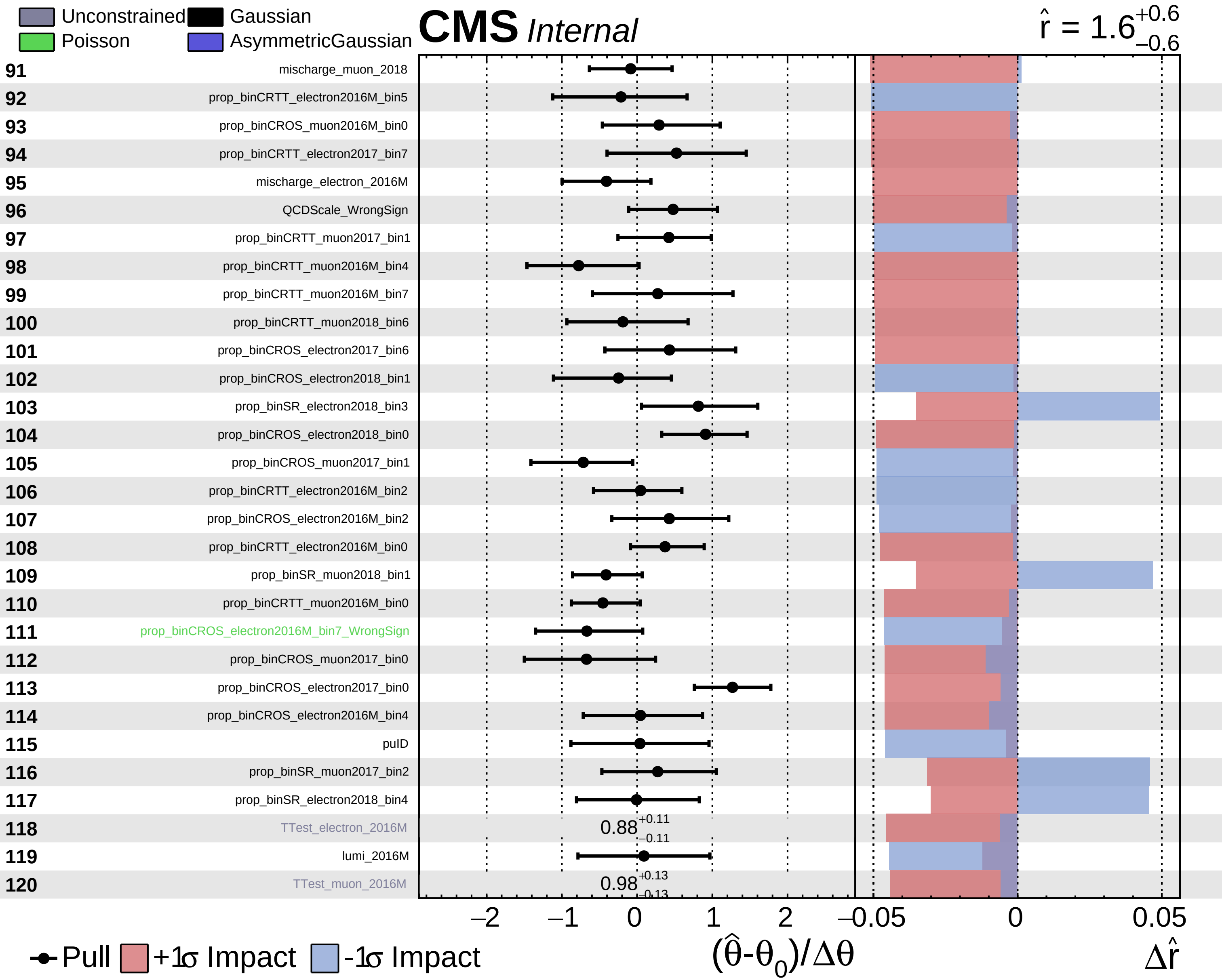


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
1 QCDScale_sig
2 pu
3 prop_binSR_muon2017_bin3
4 prop_binSR_muon2017_bin4
5 OSest_muon_2018 $0.73^{+0.30}_{-0.30}$
6 lumi_2018
7 prop_binSR_electron2016M_bin5
8 lumi_2017
9 prop_binCROS_electron2016M_bin1
10 TTest_muon_2018 $0.88^{+0.16}_{-0.16}$
11 prop_binSR_muon2018_bin5
12 prop_binSR_muon2018_bin3
13 jes_2017
14 OSest_electron_2018 $1.0^{+0.4}_{-0.4}$
15 TTest_electron_2018 $0.96^{+0.14}_{-0.14}$
16 prop_binCROS_muon2018_bin0
17 OSest_muon_2017 $1.42^{+0.33}_{-0.33}$
18 OSest_electron_2017 $0.89^{+0.34}_{-0.34}$
19 TES
20 prop_binSR_muon2016M_bin0
21 PF
22 jer_2018
23 QCDScale_TTdilep
24 prop_binSR_muon2018_bin2
25 prop_binCROS_muon2018_bin1
26 prop_binSR_electron2018_bin5
27 prop_binSR_electron2018_bin0
28 prop_binSR_muon2017_bin7
29 OSest_muon_2016M $0.8^{+0.4}_{-0.4}$
30 prop_binCROS_electron2018_bin6_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

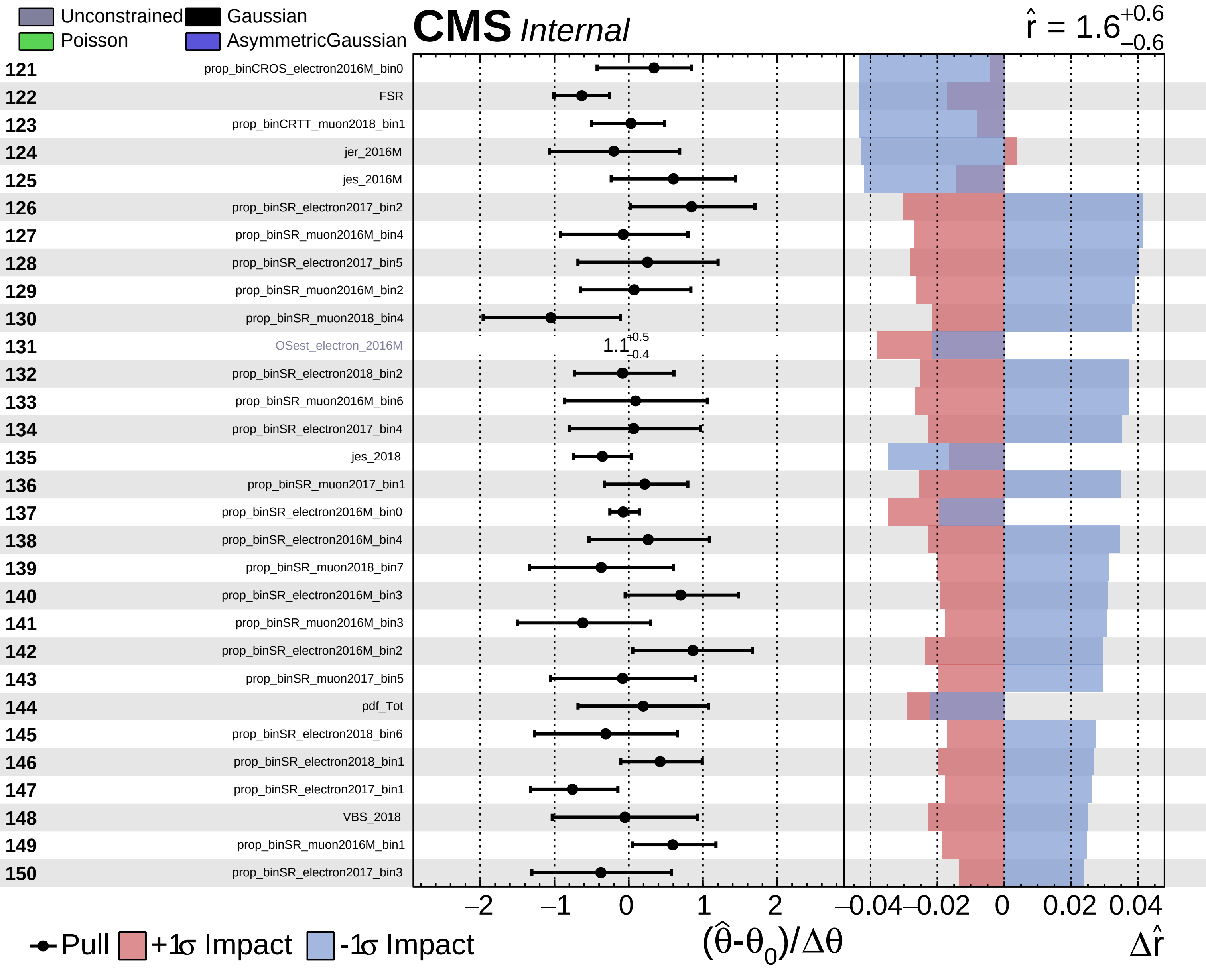


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
31 prop_binCROS_electron2017_bin4
32 prop_binCROS_electron2018_bin4
33 prop_binCROS_muon2016M_bin1
34 prop_binCRTT_muon2018_bin5
35 prop_binCRTT_muon2018_bin0
36 ISR
37 prop_binCROS_muon2018_bin4
38 prop_binCRTT_electron2017_bin1
39 prop_binCROS_electron2018_bin5
40 prop_binCROS_muon2017_bin3
41 prop_binCROS_muon2018_bin3
42 prop_binCROS_electron2017_bin1
43 TTest_electron_2017 $0.94^{+0.13}_{-0.13}$
44 prop_binCROS_muon2016M_bin3
45 prop_binCRTT_electron2018_bin0
46 prop_binCRTT_muon2016M_bin2
47 prop_binCRTT_muon2017_bin0
48 prop_binCROS_muon2016M_bin5
49 prop_binCROS_muon2016M_bin2
50 prop_binCRTT_muon2018_bin2
51 prop_binCRTT_muon2017_bin3
52 prop_binCRTT_muon2018_bin7
53 TTest_muon_2017 $1.01^{+0.11}_{-0.11}$
54 prop_binCROS_muon2017_bin6
55 prop_binCRTT_muon2018_bin3
56 prop_binCROS_electron2018_bin2
57 prop_binCRTT_electron2016M_bin4
58 prop_binCROS_muon2017_bin2
59 prop_binCRTT_electron2017_bin6
60 prop_binCRTT_electron2016M_bin3
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

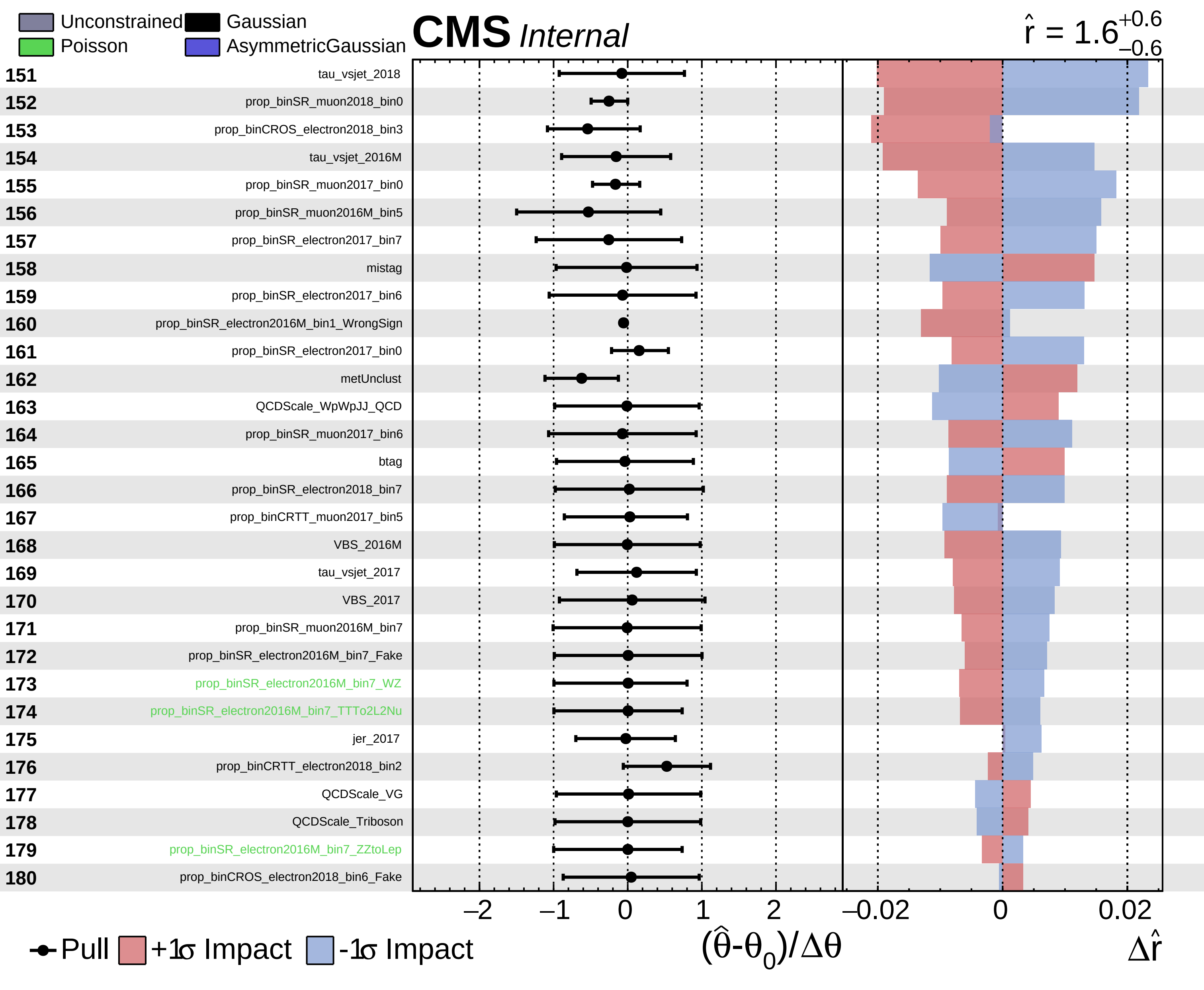

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
61 prop_binCROS_electron2016M_bin3_WrongSign
62 prop_binCRTT_electron2018_bin4
63 prop_binCROS_electron2017_bin2
64 prop_binCRTT_electron2018_bin7
65 prop_binCROS_electron2016M_bin3_Fake
66 prop_binCROS_muon2016M_bin4
67 prop_binCRTT_electron2017_bin2
68 prop_binCRTT_electron2018_bin1
69 prop_binCRTT_muon2016M_bin5
70 prop_binCRTT_electron2017_bin3
71 prop_binCRTT_muon2017_bin4
72 prop_binCROS_muon2018_bin5
73 prop_binSR_muon2018_bin6
74 prop_binCROS_electron2017_bin5
75 prop_binCROS_muon2017_bin4
76 prop_binCROS_muon2018_bin2
77 prop_binCROS_muon2016M_bin6
78 prop_binCRTT_electron2017_bin0
79 prop_binSR_electron2016M_bin1_Fake
80 mischarge_muon_2016M
81 prop_binCRTT_muon2016M_bin3
82 prop_binCROS_muon2017_bin5_WrongSign
83 prop_binCRTT_electron2017_bin4
84 prop_binCROS_electron2017_bin3
85 prop_binCROS_electron2016M_bin6
86 prop_binCRTT_muon2016M_bin6
87 prop_binCRTT_muon2017_bin2
88 prop_binCRTT_electron2016M_bin1
89 prop_binCRTT_electron2018_bin6
90 prop_binCRTT_muon2016M_bin1
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

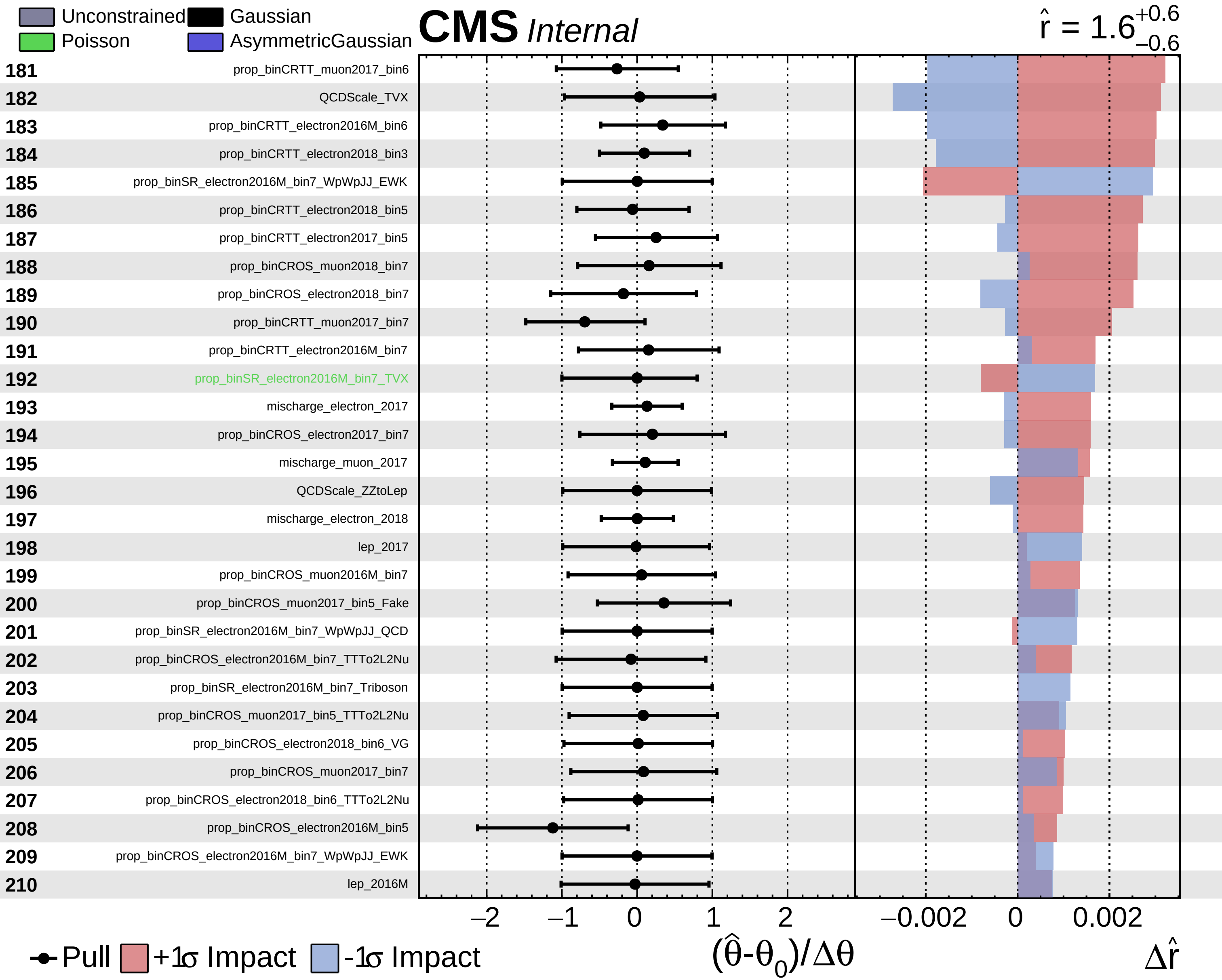


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
91 mischarge_muon_2018
92 prop_binCRTT_electron2016M_bin5
93 prop_binCROS_muon2016M_bin0
94 prop_binCRTT_electron2017_bin7
95 mischarge_electron_2016M
96 QCDScale_WrongSign
97 prop_binCRTT_muon2017_bin1
98 prop_binCRTT_muon2016M_bin4
99 prop_binCRTT_muon2016M_bin7
100 prop_binCRTT_muon2018_bin6
101 prop_binCROS_electron2017_bin6
102 prop_binCROS_electron2018_bin1
103 prop_binSR_electron2018_bin3
104 prop_binCROS_electron2018_bin0
105 prop_binCROS_muon2017_bin1
106 prop_binCRTT_electron2016M_bin2
107 prop_binCROS_electron2016M_bin2
108 prop_binCRTT_electron2016M_bin0
109 prop_binSR_muon2018_bin1
110 prop_binCRTT_muon2016M_bin0
111 prop_binCROS_electron2016M_bin7_WrongSign
112 prop_binCROS_muon2017_bin0
113 prop_binCROS_electron2017_bin0
114 prop_binCROS_electron2016M_bin4
115 puID
116 prop_binSR_muon2017_bin2
117 prop_binSR_electron2018_bin4
118 TTest_electron_2016M
$0.88^{+0.11}_{-0.11}$
119 lumi_2016M
120 TTest_muon_2016M
$0.98^{+0.13}_{-0.13}$
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

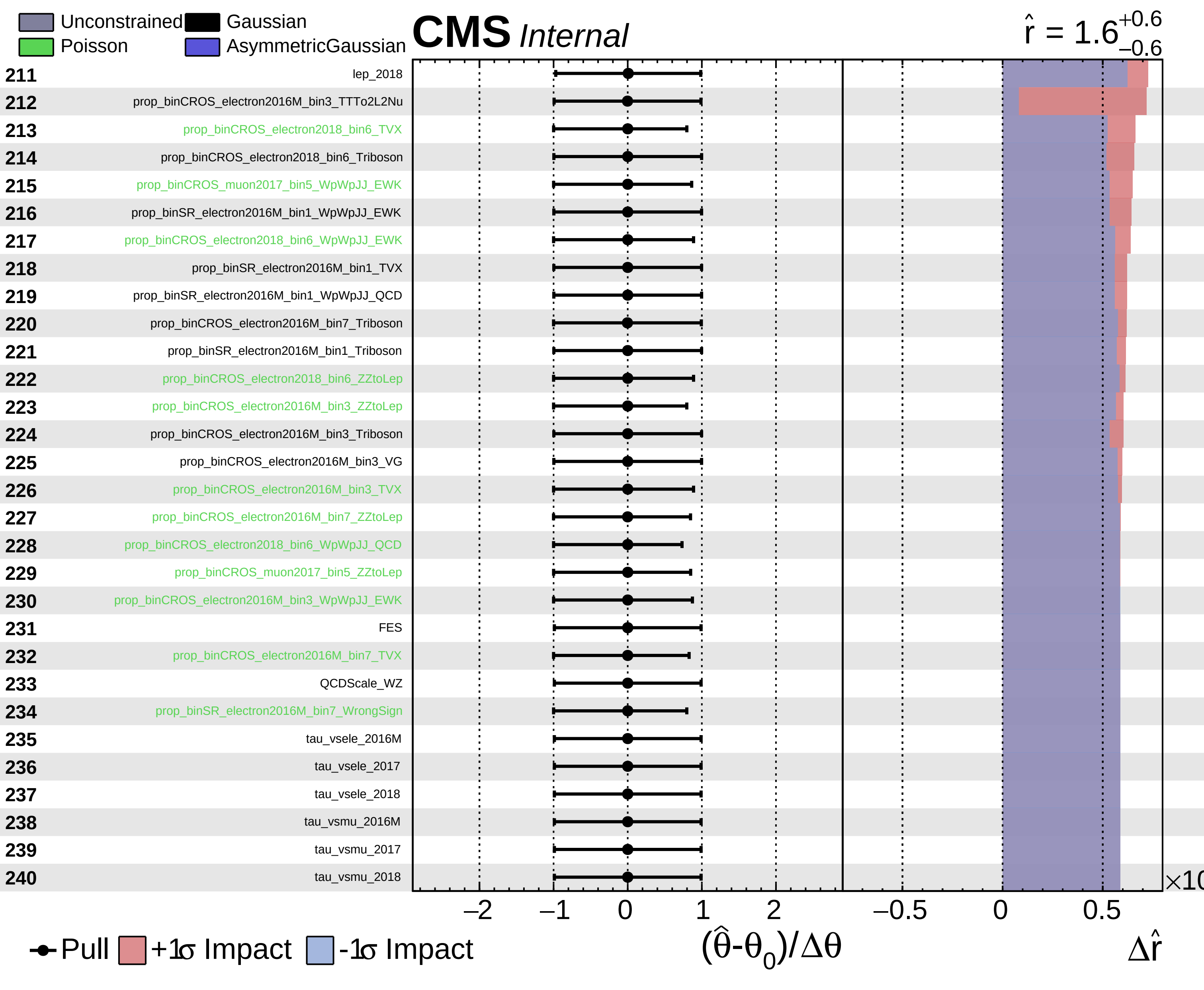


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
121 prop_binCROS_electron2016M_bin0
122 FSR
123 prop_binCRTT_muon2018_bin1
124 jer_2016M
125 jes_2016M
126 prop_binSR_electron2017_bin2
127 prop_binSR_muon2016M_bin4
128 prop_binSR_electron2017_bin5
129 prop_binSR_muon2016M_bin2
130 prop_binSR_muon2018_bin4
131 OSest_electron_2016M
$1.1^{+0.5}_{-0.4}$
132 prop_binSR_electron2018_bin2
133 prop_binSR_muon2016M_bin6
134 prop_binSR_electron2017_bin4
135 jes_2018
136 prop_binSR_muon2017_bin1
137 prop_binSR_electron2016M_bin0
138 prop_binSR_electron2016M_bin4
139 prop_binSR_muon2018_bin7
140 prop_binSR_electron2016M_bin3
141 prop_binSR_muon2016M_bin3
142 prop_binSR_electron2016M_bin2
143 prop_binSR_muon2017_bin5
144 pdf_Tot
145 prop_binSR_electron2018_bin6
146 prop_binSR_electron2018_bin1
147 prop_binSR_electron2017_bin1
148 VBS_2018
149 prop_binSR_muon2016M_bin1
150 prop_binSR_electron2017_bin3
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

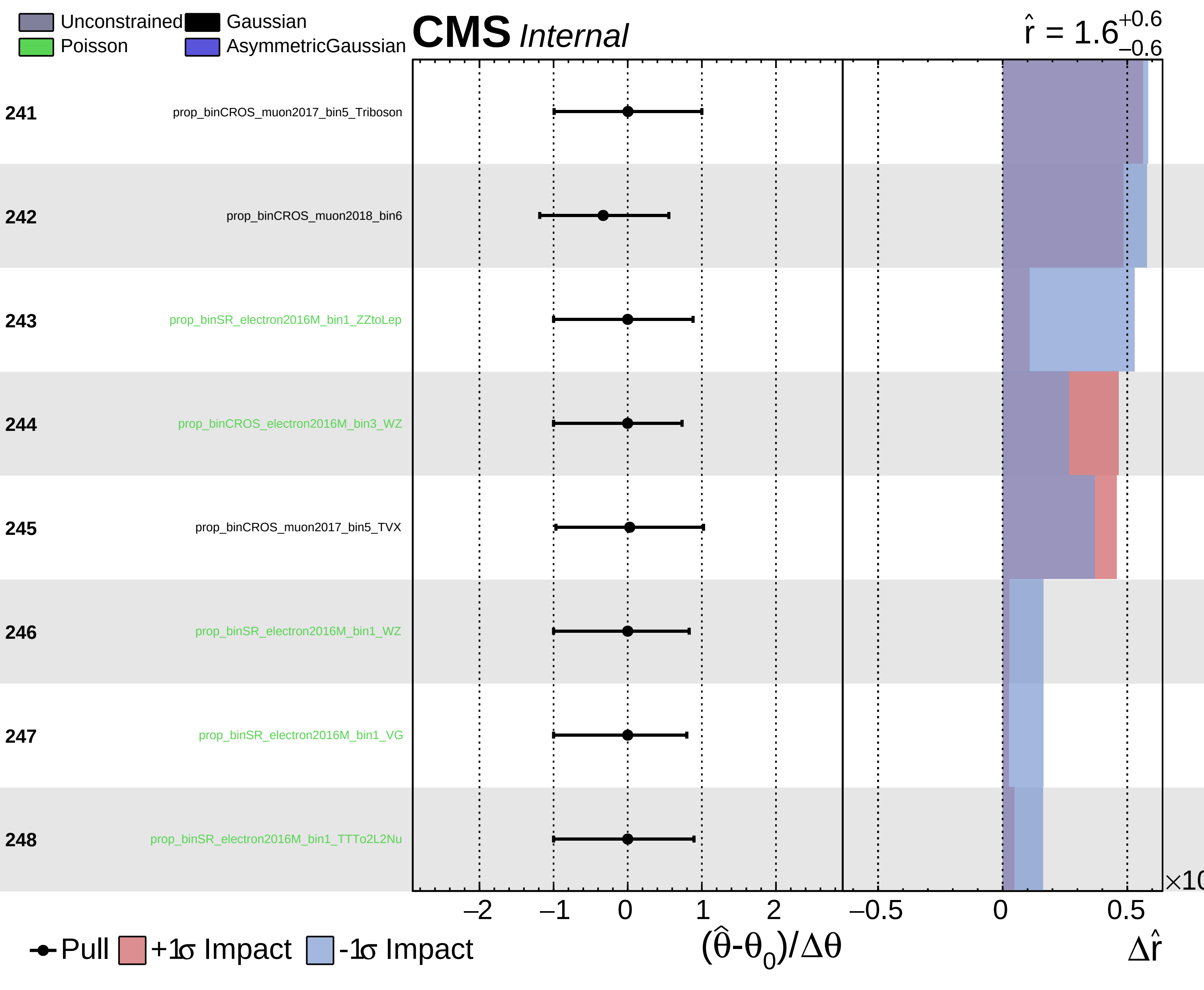

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
151 tau_vsjet_2018
152 prop_binSR_muon2018_bin0
153 prop_binCROS_electron2018_bin3
154 tau_vsjet_2016M
155 prop_binSR_muon2017_bin0
156 prop_binSR_muon2016M_bin5
157 prop_binSR_electron2017_bin7
158 mistag
159 prop_binSR_electron2017_bin6
160 prop_binSR_electron2016M_bin1_WrongSign
161 prop_binSR_electron2017_bin0
162 metUnclust
163 QCDScale_WpWpJJ_QCD
164 prop_binSR_muon2017_bin6
165 btag
166 prop_binSR_electron2018_bin7
167 prop_binCRTT_muon2017_bin5
168 VBS_2016M
169 tau_vsjet_2017
170 VBS_2017
171 prop_binSR_muon2016M_bin7
172 prop_binSR_electron2016M_bin7_Fake
173 prop_binSR_electron2016M_bin7_WZ
174 prop_binSR_electron2016M_bin7_TTTo2L2Nu
175 jer_2017
176 prop_binCRTT_electron2018_bin2
177 QCDScale_VG
178 QCDScale_Triboson
179 prop_binSR_electron2016M_bin7_ZZtoLep
180 prop_binCROS_electron2018_bin6_Fake
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.6 +0.6 −0.6
181 prop_binCRTT_muon2017_bin6
182 QCDScale_TVX
183 prop_binCRTT_electron2016M_bin6
184 prop_binCRTT_electron2018_bin3
185 prop_binSR_electron2016M_bin7_WpWpJJ_EWK
186 prop_binCRTT_electron2018_bin5
187 prop_binCRTT_electron2017_bin5
188 prop_binCROS_muon2018_bin7
189 prop_binCROS_electron2018_bin7
190 prop_binCRTT_muon2017_bin7
191 prop_binCRTT_electron2016M_bin7
192 prop_binSR_electron2016M_bin7_TVX
193 mischarge_electron_2017
194 prop_binCROS_electron2017_bin7
195 mischarge_muon_2017
196 QCDScale_ZZtoLep
197 mischarge_electron_2018
198 lep_2017
199 prop_binCROS_muon2016M_bin7
200 prop_binCROS_muon2017_bin5_Fake
201 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
202 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
203 prop_binSR_electron2016M_bin7_Triboson
204 prop_binCROS_muon2017_bin5_TTTo2L2Nu
205 prop_binCROS_electron2018_bin6_VG
206 prop_binCROS_muon2017_bin7
207 prop_binCROS_electron2018_bin6_TTTo2L2Nu
208 prop_binCROS_electron2016M_bin5
209 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
210 lep_2016M
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
211 lep_2018
212 prop_binCROS_electron2016M_bin3_TTTo2L2Nu
213 prop_binCROS_electron2018_bin6_TVX
214 prop_binCROS_electron2018_bin6_Triboson
215 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
216 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
217 prop_binCROS_electron2018_bin6_WpWpJJ_EWK
218 prop_binSR_electron2016M_bin1_TVX
219 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
220 prop_binCROS_electron2016M_bin7_Triboson
221 prop_binSR_electron2016M_bin1_Triboson
222 prop_binCROS_electron2018_bin6_ZZtoLep
223 prop_binCROS_electron2016M_bin3_ZZtoLep
224 prop_binCROS_electron2016M_bin3_Triboson
225 prop_binCROS_electron2016M_bin3_VG
226 prop_binCROS_electron2016M_bin3_TVX
227 prop_binCROS_electron2016M_bin7_ZZtoLep
228 prop_binCROS_electron2018_bin6_WpWpJJ_QCD
229 prop_binCROS_muon2017_bin5_ZZtoLep
230 prop_binCROS_electron2016M_bin3_WpWpJJ_EWK
231 FES
232 prop_binCROS_electron2016M_bin7_TVX
233 QCDScale_WZ
234 prop_binSR_electron2016M_bin7_WrongSign
235 tau_vsele_2016M
236 tau_vsele_2017
237 tau_vsele_2018
238 tau_vsmu_2016M
239 tau_vsmu_2017
240 tau_vsmu_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.6}$
241 prop_binCROS_muon2017_bin5_Triboson
242 prop_binCROS_muon2018_bin6
243 prop_binSR_electron2016M_bin1_ZZtoLep
244 prop_binCROS_electron2016M_bin3_WZ
245 prop_binCROS_muon2017_bin5_TVX
246 prop_binSR_electron2016M_bin1_WZ
247 prop_binSR_electron2016M_bin1_VG
248 prop_binSR_electron2016M_bin1_TTTo2L2Nu
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$